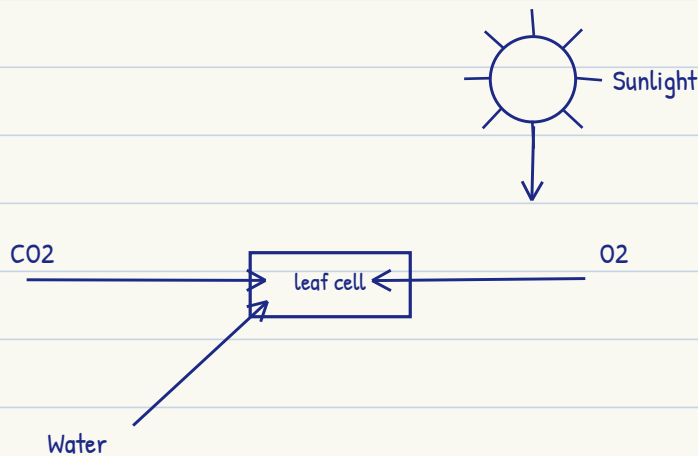


Q1

Photosynthesis is the process by which green plants prepare their own food using carbon dioxide and water in the presence of sunlight and chlorophyll, releasing oxygen as a by-product. The balanced equation is $6\text{CO}_2 + 6\text{H}_2\text{O} \xrightarrow{\text{Light/Chlorophyll}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. The raw materials are carbon dioxide and water, and the products are glucose and oxygen.

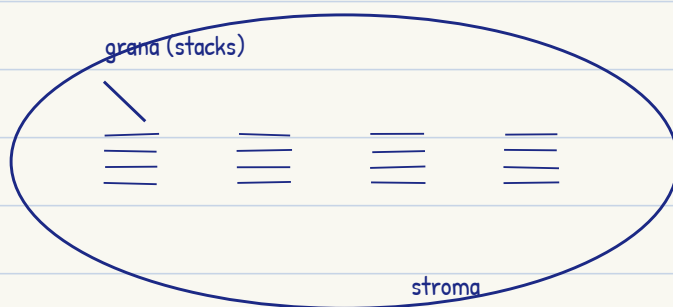


Q2

Photosynthesis occurs inside the chloroplast, a double membrane bound organelle found in large numbers in the mesophyll cells of leaves, especially in the palisade layer which is positioned close to the upper surface to receive maximum sunlight. The chloroplast is bounded by an outer membrane and an inner membrane, and between these two membranes lies a narrow intermembrane space. Inside the inner membrane is a semi fluid matrix called the stroma, which contains enzymes, ribosomes, circular DNA, and starch grains, and it is here that the dark reaction or Calvin cycle takes place.

Suspended within the stroma is an elaborate system of flattened, membrane bound sacs called thylakoids, which are stacked one above another like coins to form structures called grana, and a single chloroplast may contain anywhere from forty to sixty such grana.

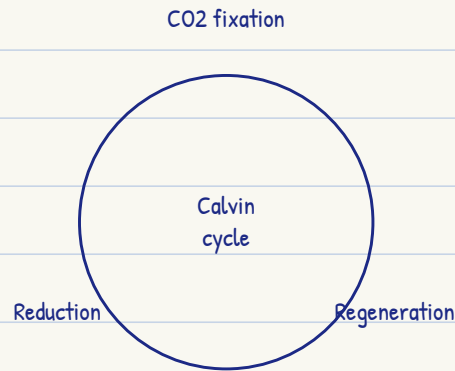
Neighbouring grana are connected to one another by thin tubular extensions of the thylakoid membrane called stroma lamellae, which help maintain a continuous compartment throughout the chloroplast and allow protons and other molecules to move between grana. The membranes of the thylakoids house the photosynthetic pigments, chlorophyll a, chlorophyll b, and carotenoids, along with the protein complexes of photosystem I and photosystem II, the electron transport chain carriers, and the enzyme ATP synthase, all of which are required for capturing light energy and converting it into chemical energy during the light reaction. Because the light reaction occurs on the thylakoid membrane within the grana, and the dark reaction occurs in the surrounding stroma, the internal architecture of the chloroplast neatly separates the two stages of photosynthesis while still allowing the ATP and NADPH produced in the grana to diffuse into the stroma where they are immediately used to fix carbon dioxide into carbohydrate.

**Q3**

Photosynthesis has two main stages. The first is the light reaction, which takes place in the grana of the chloroplast and converts light energy into chemical energy in the form of ATP and NADPH. The second is the dark reaction, which takes place in the stroma and uses this chemical energy to fix carbon dioxide into glucose.

Q4

Chlorophyll is the green pigment present in the thylakoid membranes

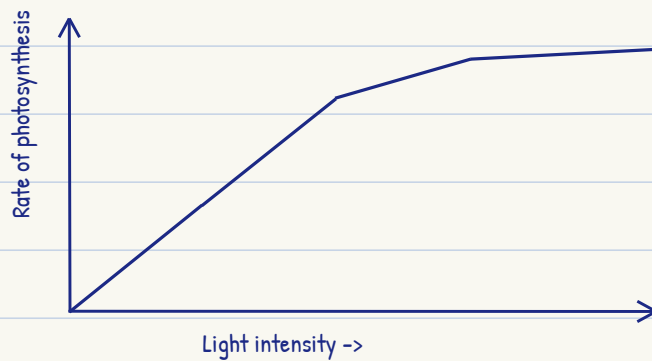
**Q7**

Stomata are tiny pores found mainly on the lower surface of leaves.

Each stoma is surrounded by two bean shaped guard cells that control its opening and closing by changing their turgidity. When guard cells absorb water and become turgid, the stoma opens, allowing carbon dioxide to enter and oxygen and water vapour to leave. When guard cells lose water and become flaccid, the stoma closes to prevent excessive water loss.

Q8

Six factors that affect the rate of photosynthesis are light intensity, carbon dioxide concentration, temperature, water availability, chlorophyll content, and the age of the leaf. As light intensity increases from low levels, the rate of photosynthesis increases almost proportionally, since more light energy is available to drive the light reaction, but beyond a certain point the rate levels off because some other factor, such as carbon dioxide concentration, becomes limiting. Temperature also has an important effect because photosynthesis depends on enzymes, which work best within an optimum temperature range, usually between twenty five and thirty five degrees Celsius for most plants; below this range enzyme activity slows down, and above it the enzymes begin to lose their shape and stop working properly, causing the

**Q9**

Take a healthy potted plant and place it in a dark cupboard for about twenty four hours so that all the starch already present in the leaves is used up. This is called destarching the plant. After this, select one leaf while it is still attached to the plant and cover the central part of it on both sides with strips of black paper or aluminium foil, leaving the rest of the leaf exposed. Now place the plant in bright sunlight for several hours. After this period, pluck the leaf and remove the covering strips. Boil the leaf in water for a minute to soften it, then boil it in alcohol using a water bath to remove the chlorophyll, which makes the leaf pale and brittle. Dip the leaf in warm water to soften it again and then place it in a dish and pour a few drops of iodine solution over it.

Observation: the parts of the leaf that were exposed to sunlight turn blue black, showing the presence of starch, while the covered part remains the original colour of iodine, showing the absence of starch. Conclusion: since starch was formed only in the part of the leaf that received sunlight, this experiment proves that sunlight is necessary for photosynthesis to occur.

Q10

Take a potted plant with a variegated leaf, that is, a leaf having both green and white patches, and destarch the plant by keeping it

light, although it depends on products of the light reaction. The light reaction produces ATP, NADPH, and oxygen, while the dark reaction uses ATP and NADPH to produce glucose. Water is split during the light reaction, whereas carbon dioxide is fixed during the dark reaction.

Q13

Photorespiration is a process in which the enzyme RuBisCO binds with oxygen instead of carbon dioxide, especially under hot and dry conditions when stomata partially close, leading to breakdown of RuBP without producing useful energy, and even resulting in loss of fixed carbon. Unlike normal respiration, photorespiration does not produce ATP, occurs only in the light, and actually lowers the efficiency of photosynthesis rather than releasing usable energy for the cell.

Q14

According to Blackman's Law of Limiting Factors, when a process is affected by several factors, the rate of the process is limited by the factor that is nearest to its minimum value at that time, regardless of the amount of the other factors present. For example, even if light intensity and temperature are optimum, if the carbon dioxide concentration in the air is low, the rate of photosynthesis will be limited by carbon dioxide alone, and increasing light further will not increase the rate until more carbon dioxide is made available. This factor is called the limiting factor at that particular time.

Q15

Photosynthesis is important for life on Earth for several reasons. Firstly, it is the primary source of food for plants, which in turn forms the base of almost all food chains and food webs. Secondly, it